A Toolbox for Differentially Private Diffusions

Refining the change-of-measure framework of Bertazzi et al. for SDE privacy in the Bayesian federated learning setting

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Background: Bayesian federated learning

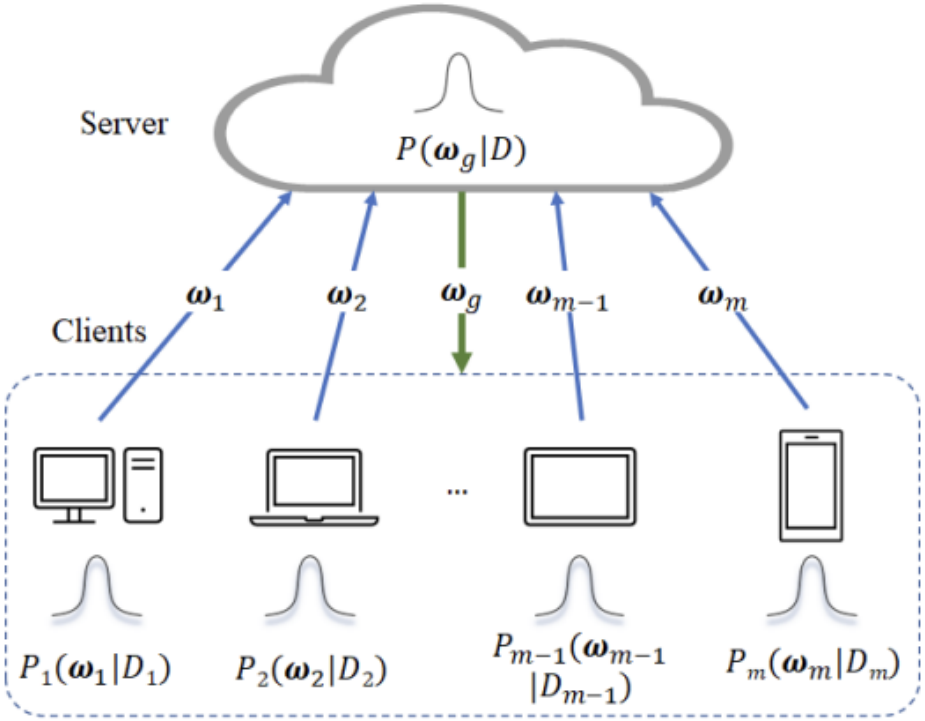

Figure 1: Cao et al. (2023)

Background: Bayesian federated learning

- Welling & Teh (2011): Stochastic Gradient Langevin Dynamics (SGLD)

$$\Delta\theta_t = \frac{\epsilon_t}{2}\left(\nabla \log p(\theta_t) + \frac{N}{n}\sum_{i=1}^{n} \nabla \log p(x_{ti}|\theta_t)\right) + \eta_t$$
$$\eta_t \sim N(0, \epsilon_t) \tag{4}$$

Figure 2: The update in SGLD, subject to $\sum_{t=1}^{\infty} \epsilon_t = \infty, \sum_{t=1}^{\infty} \epsilon_t^2 < \infty$

Background: Bayesian federated learning

- Ahn, Shahbaba, Welling (2014): Distributed SGLD (DSGLD)

$$\theta_{t+1} \leftarrow \theta_t + \frac{\epsilon_t}{2}\left\{\nabla \log p(\theta_t) + \frac{N_{s_t}}{q_{s_t}}\bar{g}(\theta_t; X^n_{s_t})\right\} + \nu_t.$$

Figure 3: The update in DSGLD, where $\bar{g}(\theta_t; X^n_{s_t}) = \frac{1}{n}_{x \in X^n_t} \nabla_{\theta_t} \log p(\theta_t; x)$ on X^n_t

- el Mekkaoui et al. (2021): Federated Stochastic Gradient Langevin Dynamics (FSGLD)
- Yang et al. (2024): Generalized Stochastic Gradient Langevin Dynamics (GSGLD)

Background: Bayesian federated learning

Methods	DP Guarantee	Scale Parameters	Parameters Updating Rule
DP-SGLD	(ϵ, δ)-DP	$\sigma = \sqrt{\frac{128NTL^2}{\tau\epsilon^2}\log\left(\frac{2.5NT}{\tau\delta}\right)\log(2/\delta)\eta_t^2} \vee \sqrt{\eta_t}$	$\Delta\theta_t \leftarrow \eta_t\left(\nabla r(\theta) + \frac{N}{\tau}\sum_{i\in S}\nabla l(x_i\|\theta)\right) + z_t$ $\theta_{t+1} \leftarrow \theta_t - \Delta\theta_t,\ z_t \sim \mathcal{N}(0, \sigma^2)$
DP-SGD	$(O(q\epsilon), \delta)$-DP	$\sigma \geq c_2 \frac{q\sqrt{T\log(1/\delta)}}{\epsilon}$	$\bar{\mathbf{g}}_t(x_i) \leftarrow Clip(\nabla_{\theta_t}\mathcal{L}(\theta_t, x_i), C)$ $\tilde{\mathbf{g}}_t(x_i) \leftarrow \frac{1}{L}\left(\sum_i \bar{\mathbf{g}}_t(x_i) + \mathcal{N}(0, \sigma^2 C^2)\right)$ $\theta_{t+1} \leftarrow \theta_t - \eta_t\tilde{\mathbf{g}}_t$
DP-GSGLD	(ϵ, δ)-DP	$\sigma = \sqrt{\eta_t C_g C_p/\epsilon}$	$\mathcal{L} \leftarrow -\sum_{i=1}^{\|\mathcal{B}_t\|}\log p(x_{t,i}\|\theta) - \log p(\theta_t)$ $\widetilde{\mathbf{g}}(\theta_t; \mathcal{B}_t) \leftarrow Clip(\nabla_{\theta_t}\mathcal{L}(\theta_t, X), C_g)$ $\Delta\widetilde{\theta}_t \leftarrow Clip(\frac{\eta_t}{2}\widetilde{\mathbf{g}}(\theta_t; \mathcal{B}_t), C_p)$ $\theta_{t+1} \leftarrow \theta_t - \Delta\widetilde{\theta}_t + \xi_t, \xi_t \sim \mathcal{N}(0, \sigma^2)$

Figure 4: Yang et al. (2024)

All introduce noising stepwise, so DP guarantees are nonuniform in time, depend on stepsize, depend on state space size etc.

If only we could understand the DP of SGLD in a more natural way!

Background: Bayesian federated learning

Bertazzi et al. (2025): can we characterize the DP of the **law** of the nth state of SGLD?

The Bertazzi et al. (2025) framework

We can create and analyze **an SDE equal in law to the Markov chain**!

Specifically, we can consider the generalized Langevin SDE

$$dX_t^D \;=\; f_D\big(X_{\kappa(t)}^D, \eta_{\kappa(t)}\big)\,dt \;+\; \sqrt{\frac{2}{\beta}}\,dW_t$$

where $X_0^D = x_0 \in \mathbb{R}^d$, $(W_t)_{t\geqslant 0}$ is a Brownian motion, $f_D : \mathbb{R}^d \times Y \to \mathbb{R}^d$ is a measurable function, $\kappa : [0,\infty) \to [0,\infty)$ is a function satisfying $\kappa(t) \leqslant t$, $(\eta_t)_{t\geqslant 0}$ is a stochastic process independent of $(W_t)_{t\geqslant 0}$ taking values in some space $\mathcal{Y}$, and $\beta > 0$ is our temperature parameter. Crucial is the introduction of $\kappa(t)$, as taking it to be the backwards projection onto the grid $\{n\gamma\}_{n\in\mathbb{N}}$ via $\kappa(t) := \gamma\lfloor t/\gamma\rfloor$ with Euler step size $\gamma > 0$ allows $(X_{n\gamma}^{\mathcal{D}})_{n\in\mathbb{N}}$ to have the same law as $(x_n^{\mathcal{D}})_{n\in\mathbb{N}}$.

The Bertazzi et al. (2025) framework

E.g., the Markov chain in the SGLD algorithm

$$x_{n+1}^{\mathcal{D}} = x_n^{\mathcal{D}} - \frac{\gamma}{s} \sum_{i \in A_{n+1}} \nabla_x \ell(x_n^{\mathcal{D}}, d_i) + \sqrt{2\gamma/\beta} z_{n+1}$$

where $A_{n+1} \subseteq [m]$ is a random minibatch of size $s \leqslant m$ and $(z_n)_{n \geqslant 1}$ is a sequence of i.i.d. standard Gaussians on R^d independent of $(A_n)_{n \geqslant 1}$ admits a continuous-time interpolation

$$dX_t^{\mathcal{D}} = -\frac{1}{s} \sum_{i \in \eta_t} \nabla_x \ell(X_{\kappa_\gamma(t)}^{\mathcal{D}}, d_i) \mathrm{d}t + \sqrt{2/\beta} \mathrm{d}W_t$$

as its SDE.

The Bertazzi et al. (2025) framework

At this point, you might ask: what is an SDE?

Definition 1.1. A Brownian Motion on $t \in [0,1]$, is a random continuous function $B : [0,1] \to \mathbb{R}$ such that:

(a) $B_t - B_s \sim \mathcal{N}(0, t-s)$ if $t \geq s$

(b) if $t_1 \leq t_2 \leq \ldots \leq t_k$, then $(B_{t_2} - B_{t_1}, B_{t_3} - B_{t_2}, \ldots, B_{t_k} - B_{t_{k-1}})$ are independent.

Figure 5: Brownian Motion

A *simple process* $H(t,\omega)$ has the form

$$H(t,\omega) = \sum_{i=1}^{k} A_i(\omega)\, \mathbf{1}_{(t_i, t_{i+1}]}(t),$$

for a partition $0 = t_0 < t_1 < \cdots < t_k = T$, where each $A_i(\omega)$ is F_{t_i}-measurable. Then we define

$$\int_0^\infty H(t,\omega)\, dB_t = \sum_{i=1}^{k} A_i(\omega) \left[B_{t_{i+1}} - B_{t_i}\right].$$

Figure 6: the Ito integral

The Bertazzi et al. (2025) framework

A typical equation is of the form

$$\mathrm{d}X_t = \mu(X_t, t)\,\mathrm{d}t + \sigma(X_t, t)\,\mathrm{d}B_t,$$

where B denotes a Wiener process (standard Brownian motion). This equation should be interpreted as an informal way of expressing the corresponding integral equation

$$X_{t+s} - X_t = \int_t^{t+s} \mu(X_u, u)\mathrm{d}u + \int_t^{t+s} \sigma(X_u, u)\,\mathrm{d}B_u.$$

Figure 7: Constructing an SDE

The Bertazzi et al. (2025) framework

How do we analyze our SDE?

What does being DP on an SDE even mean?

The Bertazzi et al. (2025) framework

Bertazzi et al.'s idea:

Let $\mathcal{A}$ be our randomized algorithm of interest. For adjacent datasets $\mathcal{D}$ and $\mathcal{D}'$, if $\mathcal{A}(\mathcal{D}')$ is defined on $(\Omega, \mathcal{F}, \mathbb{P})$, create another probability measure $\mathbb{Q}$ whose law on $\mathcal{A}(\mathcal{D})$ is equal to the law of $\mathbb{P}$ on $\mathcal{A}(\mathcal{D}')$, and try to control $\mathbb{Q}$ so that it is not that 'far' from $\mathbb{P}$.

The Bertazzi et al. (2025) framework

Naively, to create a new measure ν from μ, we can define a density f and write $\nu(E) := \int_E f \mathrm{d}\mu$ for every measurable set E. The RN theorem proves a partial converse, i.e. that any measure ν under suitable conditions can be expressed in this way with respect to μ on the same space:

Theorem 1 (Radon-Nikodym)

Let μ, ν be probability measures on a measurable space $(\Omega, \mathcal{F})$. If ν is absolutely continuous with respect to μ, i.e. $\nu(E) = 0$ for every set where $\mu(E) = 0$, then there exists a $\mathcal{F}$-measurable function $f : \Omega \to [0, \infty)$ with

$$\nu(E) = \int_E f \mathrm{d}\mu$$

for all measurable sets E. We call $\frac{\mathrm{d}\nu}{\mathrm{d}\mu} := f$ the RN-derivative.

The Bertazzi et al. (2025) framework

Since we can think of the RN-derivative as a pointwise weighting on how much more likely an outcome is under $\mathbb{Q}$ versus under $\mathbb{P}$, Bertazzi et al.'s idea is captured perfectly by the following lemma:

Lemma 2

Suppose for all adjacent $\mathcal{D}$ and $\mathcal{D}'$, there exists the measure $\mathbb{Q}$ on $(\Omega, \mathcal{F})$ specified earlier and $\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}$ is well-defined. If

$$\mathbb{P}\left(\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}} > e^{\varepsilon}\right) \leqslant \delta,$$

then $\mathcal{A}$ is (ε, δ)-DP.

Note that there is a very useful analogue for the Renyi DP case, which only requires that $\mathbb{E}[(\mathrm{d}\mathbb{P}/\mathrm{d}\mathbb{Q})^{\alpha-1}] \leqslant e^{(\alpha-1)\varepsilon}$ to guarantee (α, ε)-Renyi DP.

The Bertazzi et al. (2025) framework

How do we actually find such a $\mathbb{Q}$?

The Bertazzi et al. (2025) framework

Lemma B.1. *Consider two processes X_t^i for $i = 1, 2$ on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geqslant 0}, \mathbb{P})$ that solve the SDEs*

$$\mathrm{d}X_t^1 = g(X_{\kappa(t)}^1, \eta_{\kappa(t)})\mathrm{d}t + \sqrt{2/\beta}\mathrm{d}W_t,$$

$$\mathrm{d}X_t^2 = g(X_{\kappa(t)}^2, \eta_{\kappa(t)})\mathrm{d}t + u_t\mathrm{d}t + \sqrt{2/\beta}\mathrm{d}W_t.$$

where $X_0^1, X_0^2 = x_0 \in \mathbb{R}^d$, $(W_t)_{t\geqslant 0}$ is a standard Brownian motion, $g : \mathbb{R}^d \to \mathbb{R}^d$ is a function, $\kappa : [0, \infty) \to [0, \infty)$ satisfies $\kappa(t) \leqslant t$, $(\eta_t)_{t\geqslant 0}$ is a stochastic process independent of $(W_t)_{t\geqslant 0}$ and u_t is a $\mathcal{F}_t$-adapted process. Let the SDE given for each $i = 1, 2$ have a unique strong solution, and suppose for $T > 0$ that

$$\mathbb{E}\left[\exp\left(\frac{1}{2}\int_0^T |u_t|^2\,\mathrm{d}t\right)\right] < \infty. \tag{21}$$

Then there exists a measure $\mathbb{Q}$ such that the random variable $(X_t^1)_{t\in[0,T]}$ on the probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geqslant 0}, \mathbb{Q})$ is equal in law to $(X_t^2)_{t\in[0,T]}$ on the probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geqslant 0}, \mathbb{P})$, and furthermore

$$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = \exp\left(\sqrt{\frac{\beta}{2}}\int_0^T \langle u_s, \mathrm{d}W_s\rangle - \frac{\beta}{4}\int_0^T |u_s|^2\mathrm{d}s\right). \tag{22}$$

Figure 8: Girsanov's Theorem. Bertazzi et al. (2025)

For the above, they also assume the SDEs each have a unique strong solution, which hold for the $\kappa(t)$ we care about.

The Bertazzi et al. (2025) framework

Recall our SDEs are of the form $dX_t^{\mathcal{D}} = f_{\mathcal{D}}\big(X_{\kappa(t)}^{\mathcal{D}}, \eta_{\kappa(t)}\big)\, dt + \sqrt{\frac{2}{\beta}}\, dW_t.$

Proposition 4.6 (Privacy of the path). *Consider the family of processes (7). Suppose there exists $c > 0$ such that for every adjacent datasets $\mathcal{D}, \mathcal{D}' \in \mathcal{S}$ and $x \in \mathbb{R}^d, \eta \in \mathcal{Y}$ one has*

$$|f_{\mathcal{D}}(x,\eta) - f_{\mathcal{D}'}(x,\eta)| \leqslant c\,. \tag{10}$$

Then for $\delta > 0, \alpha \geqslant 1$ one has that the algorithm $\mathcal{A}_p(\mathcal{D}) = (X_T^{\mathcal{D}})_{t\in[0,T]}$ is $(\varepsilon_\delta, \delta)$-differentially private and $(\alpha, \varepsilon_\alpha)$-Rényi differentially private for

$$\varepsilon_\delta = C_1(T)/4 + \sqrt{C_1(T)\log(1/\delta)}, \quad \varepsilon_\alpha = \alpha C_1(T)/4\,,$$

and $C_1(T) = c^2\beta T$.

Proposition 4.5 (Privacy of the final value). *Let $T > 0$ and assume there exist constants $L, C, c > 0$ such that for every adjacent datasets $\mathcal{D}, \mathcal{D}' \in \mathcal{S}$ one has for any $x, y \in \mathbb{R}^d$ and $s \in \mathcal{Y}$*

$$|f_{\mathcal{D}}(x,s) - f_{\mathcal{D}'}(y,s)| \leqslant L|x-y| + c\,, \tag{8}$$

and $(X_t^{\mathcal{D}})_{t\in[0,T]}$ and $(X_t^{\mathcal{D}'})_{t\in[0,T]}$ are solutions of (7) with drifts $f_{\mathcal{D}}$ and $f_{\mathcal{D}'}$ respectively, driven by the same Brownian motion $(W_t)_t \geqslant 0$, such that almost surely[1] it holds that

$$\sup_{t\in[0,T]} |X_t^{\mathcal{D}} - X_t^{\mathcal{D}'}| \leqslant C\,. \tag{9}$$

Then for $\delta > 0, \alpha \geqslant 1$ one has that $\mathcal{A}_s(\mathcal{D}) = X_T^{\mathcal{D}}$ is $(\varepsilon_\delta, \delta)$-private and $(\alpha, \varepsilon_\alpha)$-Rényi private for

$$\varepsilon_\delta = C_2/4 + \sqrt{C_2\log(1/\delta)}, \quad \varepsilon_\alpha = \alpha C_2/4,$$

where $C_2 = \beta(C(L+1)+c)^2$.

Figure 9: Bertazzi et al. (2025)

The Bertazzi et al. (2025) framework

These allow Bertazzi et al. to obtain the following DP guarantees for the SGLD mechanism:

Assumption 5.5. Consider the loss function (14), where

$$\ell(x,d) = v(x,d) + k(x) \ ,$$

for functions $v : \mathbb{R}^d \times \mathcal{S} \to \mathbb{R}$ and $k : \mathbb{R}^d \to \mathbb{R}$ that are once continuously differentiable. Furthermore, ∇k is L-Lipschitz and for $c, a > 0$ one has $|\nabla_x v(x,d)| \leqslant c$ for any $x \in \mathbb{R}^d$ and any datum d, as well as

$$\langle \nabla k(x) - \nabla k(y), x - y \rangle \geqslant a\|x-y\|^2 \ .$$

We then have the following result.

Theorem 5.6. *Suppose Theorem 5.5 holds and consider the algorithm $\mathcal{A}_s(\mathcal{D}) = x_n^{\mathcal{D}}$ described in (15) with $\gamma \in (0, 2a/L^2)$ and stochastic gradient of size $s \leqslant m$. Then, for $\delta > 0$ and $\alpha \geqslant 1$, one has that $\mathcal{A}_s$ is $(\varepsilon_\delta, \delta)$-differentially private and $(\alpha, \varepsilon_\alpha)$-Rényi differentially private for*

$$\varepsilon_\delta = C_6/4 + \sqrt{C_6 \log(1/\delta)}, \quad \varepsilon_\alpha = \alpha C_6/4 \ ,$$

where $C_6 = c^2\beta(\frac{2(L+1)}{a-\gamma L^2/2} + 1)^2$.

Theorem 5.8. *Consider the loss function (14), and suppose there exists $c > 0$ such that for every $\mathcal{D}, \mathcal{D}' \in \mathcal{S}$ and $d \in \mathcal{D}, d' \in \mathcal{D}'$ one has that*

$$|\nabla_x \ell(x,d) - \nabla_x \ell(x,d')| \leqslant c.$$

Then, for $\delta > 0$ and $\alpha \geqslant 1$, the algorithm $\mathcal{A}_p(\mathcal{D}) = (x_1^{\mathcal{D}}, \ldots, x_n^{\mathcal{D}})$ shown in (15) is $(\varepsilon_\delta, \delta)$-differentially private and $(\alpha, \varepsilon_\alpha)$-Rényi differentially private for

$$\varepsilon_\delta = C_8 + \sqrt{C_8 \log(1/\delta)}, \quad \varepsilon_\alpha = \alpha C_8/4 \ ,$$

where $C_8(n) = \frac{\beta c^2}{s^2} n\gamma$.

Figure 10: Bertazzi et al. (2025)

The Bertazzi et al. (2025) framework

Where does Bertazzi et al. fall short?

Where does Bertazzi et al. (2025) fall short?

Limitation 1: Stochastic-gradient randomness is brushed aside

In the noisy SGD/SGLD section, the DP bounds are driven by *worst-case* bounds on the per-example gradients:

- Assumption 5.5: $\|\nabla_x v(x, d)\| \leqslant c$ uniformly in (x, d).
- Theorems 5.6–5.8 then bound privacy in terms of this global constant (plus Lipschitz/strong convexity of k), with only very weak dependence on minibatch size s and dataset size m.

Where does Bertazzi et al. (2025) fall short?

The results of Ryffel et al. (2022) and Ye & Shokri (2022) suggest that one may obtain superior bounds by fully exploiting the randomness of the stochastic gradient. However, for simplicity we do not consider this in the current work.

Figure 11: Bertazzi et al. (2025)

While assuming be acceptable for a first pass in the MCMC setting, in Bayesian federated learning, the randomness of the stochastic gradient is not a minor detail, and is instead the main privacy-saving structure!

Where does Bertazzi et al. (2025) fall short?

Limitation 2: Sup-type stability assumptions are too rigid

Their Girsanov-based RN bounds rely on *deterministic* sup-type control:

- Path-level condition (for the final draw):

$$\sup_{t\in[0,T]} \|X_t^D - X_t^{D'}\| \leqslant C \quad \text{a.s.}$$

  together with a uniform drift bound $|f_D(x,\eta) - f_{D'}(x,\eta)| \leqslant c$ for all x, η.
- In the SGLD specialization, this is enforced via global bounds on $\|\nabla_x \ell(x,d)\|$ and a strongly convex "regularizer" $k(x)$.

These assumptions are:

- Very strong in high dimension (they control the *worst* possible trajectory for the *worst* datum).
- Blind to typical behaviour of the stochastic gradient and to how often a specific user's data is actually used.

Contributions

Bertazzi et al.: $\sup_{t\in[0,T]} |X_t^{\mathcal{D}} - X_t^{\mathcal{D}'}| \leqslant C$ a.s.

More realistic:

Assumption 0.1 (High-probability difference bound)

$\forall\ T \geqslant 1$ we have $c_T > 0$ and $\delta'_T \in [0,1)$ independent of adjacent $\mathcal{D}, \mathcal{D}'$ where

$$\mathbb{P}[\sup_{0\leqslant t\leqslant T} |X_t^{\mathcal{D}} - X_t^{\mathcal{D}'}| > c_T] \leqslant \delta'_T$$

Contributions

Theorem 3 (Relaxed privacy for the final value)

Say 0.1 holds. Then for any $T \geqslant 1$ and any $\delta \in (\delta'_T, 1)$, the mechanism A_s is $(\varepsilon_T(\delta), \delta)$-differentially private, with

$$\varepsilon_T(\delta) = \frac{C_T}{4} + \sqrt{C_T \log\Big(\frac{1}{\delta - \delta'_T}\Big)},$$

where

$$C_T := \beta\big(c_T(L+1) + c\big)^2$$

Contributions

Proof sketch:

- Define good event $G_T = \left\{ \sup_{0 \leqslant t \leqslant T} \left| X_t^{\mathcal{D}} - X_t^{\mathcal{D}'} \right| \leqslant c_T \right\}$. We have

$$\mathbb{P}\Big[\frac{dP}{dQ} > e^{\varepsilon_T(\delta)}\Big] \leqslant \mathbb{P}(G_T^c) + \mathbb{P}\Big[\frac{dP}{dQ} > e^{\varepsilon_T(\delta)} \,\Big|\, G_T\Big]$$

- In the latter case, our behavior is nicely bounded, so we suitably shift and apply Girsanov's theorem, etc. By taking a wider bound on the RN derivative we shrink the tail to accommodate for $\mathbb{P}(G_T^c) = \delta'_T$

Contributions

Theorem 4

Consider the SGLD markov chain defined in Bertazzi (15):

$$x_{n+1}^{\mathcal{D}} = x_n^{\mathcal{D}} - \frac{\gamma}{s} \sum_{i \in A_{n+1}} \nabla_x \ell(x_n^{\mathcal{D}}, d_i) + \sqrt{2\gamma/\beta} z_{n+1}.$$

Suppose Bertazzi 5.5 holds, and consider the algorithm $\mathcal{A}_s(\mathcal{D}) = x_n^{\mathcal{D}}$ with $\gamma \in (0, 2a/L^2)$ and $s \leqslant m$. Then, for $\alpha \geqslant 1$, one has that $\mathcal{A}_s$ is $(\alpha, \varepsilon_\alpha)$-Renyi DP with

$$\varepsilon_\alpha := \frac{N}{\alpha - 1} \log\left(1 - s/m + (s/m)e^{\tau}\right), \quad \tau := \beta\alpha(\alpha - 1)c^2\gamma/s^2.$$

Contributions

Proposition 0.1

The SDE associated to DSGLD is

$$d\Theta_\tau = \frac{1}{2}\left(\nabla \log p(\Theta_\tau) + \frac{N_{S_\tau}}{q_{S_\tau}}\,\bar{g}(\Theta_\tau; X_{S_\tau}^{n_{S_\tau}})\right) d\tau + dW_\tau$$

where for our drift $b_{\mathcal{D}}(\Theta_\tau) = \frac{1}{2}\left(\nabla \log p(\Theta_\tau) + \frac{N_{S_\tau}}{q_{S_\tau}}\,\bar{g}(\Theta_\tau; \cdot)\right)$ we have

$$b_{\mathcal{D}'}(x) - b_{\mathcal{D}}(x) = u_\tau(x) = \begin{cases} 0, & S_\tau \neq s^*, \\ \dfrac{N_{s^*}}{q_{s^*}}\left(\bar{g}'(x) - \bar{g}(x)\right), & S_\tau = s^* \end{cases}$$

This is **random** and **not absolutely continuous**, dependent on how we randomly sample our shards!

Contributions

We want to show DP for DSGLD, but we don't meet the conditions for Bertazzi's method. How can we circumvent this?

(Possible) Proof sketch: Fix a realization of the shard selection $S_\cdot$. Conditioning on $S_\cdot$ we have

$$d\Theta_\tau = b_{\mathcal{D}}\big(\Theta_\tau, S_\tau(\omega)\big)\, d\tau + dW_\tau$$

then $u_\tau(\Theta_\tau, S_\tau) = b_{\mathcal{D}'}\big(\Theta_\tau, S_\tau\big) - b_{\mathcal{D}}\big(\Theta_\tau, S_\tau\big)$ is deterministic, which we w.t.s. is bounded. Then by Girsanov's we have

$$\frac{d\mathbb{Q}^S}{d\mathbb{P}^S} = \exp\left(\int_0^T \big\langle u_\tau, dW_\tau \big\rangle - \frac{1}{2}\int_0^T \|u_\tau\|^2\, d\tau\right)$$

where we w.t.s. we can take $\mathbb{Q}$ a composition of our $\mathbb{Q}^S$ to get a proxy for the RN-derivative, which we then use for DP.

Next steps

- Flesh out the proof of DP for DSGLD
- Extend DP to FSGLD and other mechanisms derivative of DSGLD
  - This may assume require further relaxations that we will show
- Ideas?